Testing Portfolio Exposure from Text with Matched Holdings

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Abstract

A correct portfolio total can reveal little about how an account description was aggregated. When all products hold the same percentage of a target asset, any normalized weighting of the products gives the correct exposure. We construct matched questions that hold the true account exposure fixed while changing which product weights matter. On 24 such pairs, a greedy Qwen3-8B configuration answered all 24 homogeneous questions correctly and 4 heterogeneous questions correctly. A reasoning-enabled configuration of the same quantized checkpoint answered all 48 correctly. On 26 ledger requests derived from published fund tables, the original configuration reported correct fields with a wrong total 18 times. Recomputing its reported fields recovered all 18 totals. Across 100 paired prompts, final-total accuracy rose from 35 to 98 correct responses. Both remaining failures exhausted the output budget on the same split account. These findings show which questions distinguish the tested configurations and how a reported ledger localizes a discrepancy between source fields and an aggregate.

1 What does a correct total tell us?

An investor puts GBP 2,500, 1,500, 4,000 and 2,000 into four products. Suppose each product holds 55% of Asset A. A system that reads every position, one that drops the largest, and one that weights the four products equally can all report 55%. The answer alone supplies no evidence that the four investment amounts were used correctly.

Table 1 gives the first family in our evaluation. Keeping the amounts and the true total fixed, we change the four product percentages. Equal product weighting now yields 46.75%, more than eight percentage points below the truth. On this exact text, the original model output was 44.44%. That number differs from both the correct answer and the equal-weight shortcut. The example separates a property of the question from a hypothesis about the model's internal calculation.

	Product 1	Product 2	Product 3	Product 4	True total
Position value (GBP)	2,500	1,500	4,000	2,000	10,000
Same target allocation (%)	55	55	55	55	55
Varied target allocations (%)	67	27	78	15	55

Table 1. A matched family. Qwen's original direct answers were 55% and 44.44%, respectively. The recheck returned 55% for each.

The paper asks when a portfolio question can reveal a weighting error, and which part of an answer should be inspected when the total is wrong. We first specify the accounting relationship and the counterfactual test. We then compare the same model on the same questions under two inference configurations. A separate ledger score records whether a discrepancy is already present in the extracted holdings or appears when the model reports the total.

2 Design and scoring

Let v_i be the pounds invested in funded lot i, and let a_i be the percentage of its parent product allocated to the named target. The account-level percentage is

$$\theta = \sum_i q_i a_i, \qquad q_i = v_i / \sum_j v_j. \tag{1}$$

For any alternative nonnegative weights p_i summing to one, the error is $\sum_i (p_i - q_i) a_i$. If every $a_i = c$, the error is zero. Varying a_i can expose a particular weight error, although one heterogeneous vector can still be insensitive to some errors. The supplement derives the independent contrasts needed to identify a fixed linear weighting rule and shows the limits of applying that result to flexible model responses.

Controlled accounts. Each of 24 designed families contains four unequal investment amounts. One question uses identical product allocations and the paired question uses four distinct allocations with the same correct total. In each heterogeneous case an unweighted product mean misses the true answer by at least five percentage points. The recheck includes all 48 merged direct-answer questions. The earlier Study B also split a position across repeated lots, requested ledgers and changed the economic exposure. Those additional controlled conditions retain their original results in the supplement.

Published holdings. We use 88 holding-name and percentage rows from eight funds in four Vanguard and BlackRock documents dated September 2026 [1, 6]. We form all ten within-Vanguard fund pairs and all three within-BlackRock pairs. Each constructed GBP 10,000 account has two presentations, one with a position split into separately funded lots. Each presentation is queried twice, once for a total alone and once for a per-lot ledger and total. This gives $13 \times 2 \times 2 = 52$ prompts. Tables were transcribed and standardized before inference. In each prompt the target is a named underlying fund, and absence of that name from the supplied table counts as zero. The outcome measures look-through exposure to that instrument. It does not measure future volatility.

Paired configuration comparison. We reused the saved original responses to all 100 inputs and collected one new response per input. Both configurations use the same pinned Qwen3-8B Q4_K_M checkpoint [4], system and user text, and JSON-object constraint. The original used greedy decoding, disabled thinking and a 512-token output cap. The second enabled thinking and used temperature 0.6, top-p 0.95, top-k 20, min-p 0 and up to 4,096 output tokens. The seed and runtime context/batching also changed. The recheck protocol was frozen before its evaluation calls, after we had examined the original answers. Four engineering calls on separate prompts are excluded. Each condition contributes one response per prompt.

Scoring. The primary score uses the requested JSON and the final answer field. A total must lie within 0.5 percentage points of the accounting value. A correct ledger must contain exactly the expected lot IDs, each amount within GBP 0.005 and each allocation within 0.00005 percentage points. An external calculator applies Equation 1 to the amounts and percentages *reported by the model*. No gold values enter this calculation. Responses cut off at the token limit count as failures. A secondary format-only reading gives the same scores on this recheck. The 100 prompts reuse 24 families and 13 accounts. All rates below describe this fixed set.

3 How the same questions score under two configurations

Outcome	Original	Reasoning enabled
Identical allocations, correct direct total	24/24	24/24
Varied allocations, correct direct total	4/24	24/24
Published tables, correct direct total	2/26	25/26
Published tables, correct total alongside ledger	5/26	25/26
Correct final totals, all prompts	**35/100**	**98/100**
Published tables, all reported fields correct	23/26	25/26
Calculator correct using those fields	23/26	25/26
Correct fields alongside wrong model total	18/26	0/26

Table 2. Strict, paired scores. The bottom three rows describe the same 26 ledger prompts as the fourth row. They are excluded from the 100-prompt sum.

The homogeneous-minus-heterogeneous accuracy gap was 83.3 percentage points under the original configuration. A percentile bootstrap resampling the 24 complete families gave a descriptive 95% range of 66.7 to 95.8 points. On recheck both groups reached the ceiling. Their observed gap was zero and its empirical bootstrap distribution was degenerate. Population equivalence would require evidence beyond these families. At the individual-prompt level, 64 previous failures became correct, 34 were correct both times, one previously correct answer failed and one failed both times.

Both remaining failures. The two unsuccessful responses came from the split version of the same constructed BlackRock MyMap 4/MyMap 5 account. The task asked first for a direct total and separately for a ledger. Each response consumed all 4,096 output tokens and ended with an empty final answer. The two parent-fund target percentages were 17.89% and 17.88%, so every convex average of them lies within the 0.5-point scoring tolerance. Here the measurable failure was producing the required final answer within budget. This single account provides too little variation to attribute the timeouts to splitting.

Relation to the wider experiment. In the earlier greedy Study B, both Qwen and Phi-4 14B scored 48/48 on homogeneous direct prompts across merged and split presentations. They scored 5/48 and 7/48, respectively, on heterogeneous direct prompts. The 4/24 figure in Table 2 uses only the Qwen merged subset. The reasoning-enabled recheck covers Qwen only. It supplies a configuration comparison within one checkpoint and leaves the corresponding Phi and controlled split-position comparisons open.

4 Locating an error in a published-table answer

Two recorded accounts show why the same wrong total can call for different corrections. Both use the Vanguard U.S. Equity Index Fund as the target. Every number below comes from a saved ledger response or from recomputing that response.

Correct fields, wrong total. A constructed account invests GBP 6,900 in LifeStrategy Global 40% Equity and GBP 3,100 in Global 80% Equity. The supplied tables give target weights of 6.8% and 28.3%. Both configurations reported those four numbers correctly. The original response reported a total of 14.6%. Its own rows imply

$$0.69(6.8) + 0.31(28.3) = 13.465\%.$$

The reasoning-enabled response reported 13.465%. Across the 26 original ledger prompts there were 18 correct ledgers with an incorrect reported total. Recomputing the reported rows recovered the correct total in all 18. This identifies an inconsistency in observable output. The ledger does not reveal the model's private computational procedure.

Wrong allocation attached to a lot. Another account splits a Global 20% position into GBP 1,860 and GBP 1,240 and invests GBP 6,900 in Global 60%. Its three target percentages should be $(0, 0, 14.8)$. The original answer reported $(14.8, 14.8, 14.8)$ and a 14.8% total. Correct arithmetic on those rows also gives 14.8%. The reference is 10.212%. The recheck reported the three source percentages and the total correctly. The original source run contained three field errors, all on split accounts involving the supplied zero-target table. Their shared inputs limit any inference about the frequency or cause of such errors.

Comparison with a parser. A format-specific parser in Study B solved all 52 published-table prompts. These standardized texts admit an ordinary extraction-and-calculation solution. The comparison matters for evaluation because the ledger allows a discrepancy to be assigned to reported fields, arithmetic consistency or the output interface. Any claim about extracting from raw PDFs would require a document-level experiment and independently checked annotations.

5 Interpretation and scope

Matched composition and textual fragmentation test different properties. Changing product allocations while holding the account total fixed can reveal sensitivity to invested amounts. Splitting a funded lot while holding both holdings and exposure fixed tests invariance to presentation. The original Qwen heterogeneous direct score fell from 4/24 merged to 1/24 split, while Phi's rose from 2/24 to 5/24. The two configurations gave opposite directions. In the 100-prompt recheck the clearest result is configuration dependence on the merged heterogeneous cases.

Relation to prior methods. Behavioural testing in NLP uses controlled input changes to expose failures that aggregate accuracy can hide [3, 5]. Interpreter-based methods separate extraction or program generation from execution [2]. We use ordinary weighted averaging and a calculator for these established purposes. The portfolio-specific contribution is to choose paired questions with the same gold exposure and different sensitivity to weighting, then retain enough reported fields to diagnose individual errors. A fuller comparison with financial QA, presentation effects and numerical verification appears in the supplement.

What the earlier diagnostic contributes. An eight-query acceptance rule from Study A accepted zero cases for either tested configuration. An exact-checkable convex feasibility analysis could exclude a fixed normalized weighting explanation for recorded traces, with only one extra Qwen exclusion beyond the specified simple witnesses. The supplement retains the proofs, rejected cases and counterexamples. The empirical contribution in the main text rests on the paired questions and the ledger observations. A working correctness certificate remains an open problem.

Limits and next test. The accounts were constructed, the issuer tables were internally transcribed with AI assistance, and the recheck used previously examined inputs. The final 31 responses ran on a replacement CPU host after verifying the model and inference-program hashes. Reusing the same checkpoint removes variation in weights, while the joint changes to sampling, reasoning, budget and execution preclude isolating any single cause. A further test could freeze fresh issuer tables, obtain independent human transcriptions and vary reasoning and output budget separately while holding sampling fixed where supported. Repeated draws would address run-to-run variability.

Conclusion. A portfolio total is informative about weighting only to the extent that plausible weighting errors would change it. Matched heterogeneous holdings exposed a difference of 20 correct answers between two configurations that both passed every homogeneous control. On the published tables, the ledger supplied a second check: the earlier run often copied the input fields faithfully yet reported a total inconsistent with them. Together these tests specify what the observed score establishes and what should be inspected when a portfolio answer fails.

Data and disclosure. The companion package includes the registered prompts, original and recheck replies, scores, failure census and source manifests. Study A's risk calculations use supplied return scenarios. The main endpoint is exposure to a named asset or underlying fund. AI assistance was used for design, implementation, internal source checks and drafting.

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Technical supplement

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The main paper reports the 100-prompt configuration comparison. This supplement develops the fixed-weight identification argument, documents the earlier acceptance policy, and retains the original controlled and source-table experiments. The three studies have different experimental units. Study A evaluates exposure and risk diagnostics on traces; Study B evaluates composition and presentation; Study C reuses a specified subset of Study B inputs to compare Qwen inference configurations. Its strict results appear in the main paper.

A Evidence map and chronology

The argument uses three distinct types of evidence: analytical statements under explicit assumptions, deterministic injected-error checks, and saved model responses. An artificial faulty rule that passes a test is a counterexample to the test's identifying power, not an observation that a model used that rule. A reported ledger is an output, not a transcript of hidden computation.

The empirical sequence is also explicit. A 472-call pilot motivated the design but contained an illustrative-number confound. Study A introduced cue-free tasks, an eight-query diagnostic, held-out composition challenges and issuer-derived holdings. Its failed acceptance policy and restricted explanation class motivated Study B's simpler matched transfer and ledger analysis. Study B's model replication is prospective relative to its locked prompts, but it follows knowledge of the earlier study. Neither study is externally preregistered.

Evidence	Cases / materials	Saved calls	Role
Pilot	16 initial economic cases; smaller follow-ups	472	Motivation and confounds; Appendix L
Study A core	16 evaluation families, 32 cases per model	576 per model	Contrast audit, consistency, exposure and risk
Study A extensions	20 source accounts; 2 new compositions per core case	60 source + 64 challenge per model	Source transfer and unseen-composition diagnosis
Study A execution	Runtime-based selection; not new cases	167 rechecks + 8 development	Separate execution amendment / development
Study B controlled	24 fresh matched families	216 per model	Homogeneous versus heterogeneous transfer and ledgers
Study B sources	13 accounts; 8 funds, 4 PDFs, 2 issuers	52 per model	Interface, fields and totals on published values
Study C	48 controlled + 52 source prompts from Study B	100 new Qwen responses	Paired configuration comparison; 4 development responses excluded

Table 1: Study A uses SmolLM3-3B and Qwen3-8B; Study B uses Qwen3-8B and Phi-4 14B. Calls, cases, families and documents are distinct denominators. The table is a provenance map, not a pooled sample size.

The two source tiers are different. Study A uses five Vanguard LifeStrategy factsheets and three named target funds. Study B uses a Vanguard LifeStrategy *Global* report and BlackRock MyMap factsheets with one named target per publisher. Source documents, target definitions and tolerances differ; their success rates cannot be compared as an intervention effect. All accounts

are constructed. Authentic published holdings do not make the account allocations real customer data.

Study C froze the prompts, scoring and configuration before its evaluation run. Earlier responses on these same questions were already known. Each of the 100 new responses is paired with a saved Study B response. The 24 controlled families enter Study C only in their merged, direct-answer form.

B Accounting, risk, and what a correct answer identifies

Let $q_j \geq 0$ be the account fraction allocated to product j, with $\sum_j q_j = 1$. Let B_{ja} be the fraction of that product in underlying asset a. The aggregate exposure, state return, and standard deviation are

$$w = B^\top q, \qquad R_s = w^\top r_s, \tag{1}$$

$$\sigma(w) = \sqrt{\sum_s \pi_s (R_s - \mu)^2} = \sqrt{w^\top \Sigma w}, \qquad \mu = \sum_s \pi_s R_s. \tag{2}$$

Equation 1 is ordinary weighted accounting; Equation 2 is classical portfolio mathematics [15]. All comparisons fix the joint return distribution and exclude fees, taxes, leverage, trading, tracking error, counterparty risk, and other wrapper-specific effects. These assumptions define a controlled task rather than a claim that actual financial wrappers are economically interchangeable.

Risk accuracy is also weaker than exposure identification. For two assets and $D = R_A - R_B$, classical variance algebra gives

$$\sigma^2(a) = \mathrm{Var}(R_B) + 2a\,\mathrm{Cov}(R_B, D) + a^2\,\mathrm{Var}(D). \tag{3}$$

If $\mathrm{Var}(D) > 0$ and the minimum $a_* = -\mathrm{Cov}(R_B, D)/\mathrm{Var}(D)$ lies inside $[0, 1]$, distinct admissible allocations equidistant from a_* can have exactly the same risk. When $\mathrm{Var}(D) = 0$, all allocations have the same standard deviation. These are standard consequences of the variance formula, not new risk theory. They explain why a correct risk number alone cannot generally certify correct extraction; tolerance-based scoring can obscure further exposure errors.

For two assets, write $b_j = B_{jA}$ and $B_{jB} = 1 - b_j$. The correct account fraction in A is $a = q^\top b$. Consider the restricted error class

$$f_m(b) = \frac{\sum_j m_j q_j b_j}{\sum_j m_j q_j} = p^\top b, \qquad p_j = \frac{m_j q_j}{\sum_k m_k q_k}, \tag{4}$$

where $m_j \geq 0$ and the denominator is positive. Omitting a product corresponds to $m_j = 0$; duplicating it corresponds to $m_j = 2$. More generally m_j represents a relative weighting distortion. We introduce this class as an analytical model, not as an assertion about a language model's hidden computation.

Proposition 1 (Homogeneous-holdings blind spot)**.** *If $b_j = a_0$ for every product, then $f_m(b) = a_0$ for every admissible multiplicity vector m. Neither aggregate exposure nor any risk statistic computed solely from that exposure distinguishes these errors from correct aggregation.*

Proof. Since $\sum_j p_j = 1$, $p^\top(a_0 \mathbf{1}) = a_0$. The aggregate B exposure is $1 - a_0$, so every state payoff and every statistic of those fixed payoffs also coincides. □

The result covers all choices of a_0, not just balanced portfolios. Testing many homogeneous product mixtures at different stock/bond proportions still cannot identify product-specific weighting

distortions. Uniform multiplication of all m_j is economically irrelevant after normalization and should not be treated as an exposure error.

For a single heterogeneous composition b, a wrong p may still satisfy $(p-q)^\top b = 0$. Thus correctness on one heterogeneous instance is also insufficient in general. For interior account weights and nonconstant b, normalization and one exact aggregate observation leave, generically, $J-2$ unconstrained weighting directions. This is an elementary rank observation, not a new result in linear algebra.

C Diagnosing aggregation with informative interventions

C.1 Interventions and conditional identification

The blind spot suggests a design requirement: a useful probe must make the reported total depend on the weights whose correctness is being assessed. Reordering or splitting identical holdings cannot do this for the error class in Equation 4. Changing product compositions can. We derive this diagnostic under an explicit fixed-rule assumption; Study A subsequently tests that assumption and the acceptance rule rather than assuming they hold for an LLM. Hold account weights, product labels, and sentence structure fixed. For identification, let J count positively funded products; zero-weight products can be removed. For contrast vector c_ℓ, modify each product's A allocation by $\pm\delta c_{\ell j}$ and its B allocation by the opposite amount. Choose δ so every modified allocation remains in $[0,1]$. These interventions deliberately change economic exposure; they complement the payoff-preserving fragmentation tests rather than pretending to be additional invariant rewrites.

For an observed aggregate-exposure function f, define

$$g_\ell = \frac{f(b+\delta c_\ell) - f(b-\delta c_\ell)}{2\delta}, \qquad s = g - Cq. \tag{5}$$

Under Equation 4 with *the same* p across interventions, $g = Cp$. In particular, the expected response depends on the account allocations q, not the unknown base aggregate answer.

Proposition 2 (Identification from contrast measurements)**.** *Within the class of fixed normalized linear aggregation rules $f(b) = p^\top b$, the contrast measurements $g = Cp$ uniquely identify every admissible weight vector p if and only if*

$$\operatorname{rank}\begin{bmatrix}\mathbf{1}^\top \\ C\end{bmatrix} = J. \tag{6}$$

Consequently, identification from contrasts and normalization requires at least $J-1$ independent contrast measurements.

Proof. If two normalized vectors produce the same measurements, their difference e satisfies $\mathbf{1}^\top e = 0$ and $Ce = 0$. Full column rank forces $e = 0$. If rank is deficient, take a nonzero null vector e. Around any interior normalized weight vector, a sufficiently small nonzero multiple of e preserves nonnegativity and gives an indistinguishable alternative. The row count gives the lower bound. □

The lower bound is scoped to global identification within this observation scheme. A particular boundary vector can be uniquely identified with fewer measurements because of nonnegativity. The proposition does not assert optimality among arbitrary algorithms supplied with different information or additional ground-truth answers.

For the four-product experiment, use the three nonconstant rows of a Hadamard matrix:

$$C = \begin{bmatrix} 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}, \quad C^\top C = 4I - \mathbf{1}\mathbf{1}^\top. \tag{7}$$

The effective weighting error and weights have the closed forms

$$\widehat{e} = \tfrac{1}{4} C^\top s, \qquad \widehat{p} = q + \widehat{e}. \tag{8}$$

These identify normalized relative weights, not the absolute scale of multiplicities. Even perfect recovery of p cannot distinguish m from a positive scalar multiple of m.

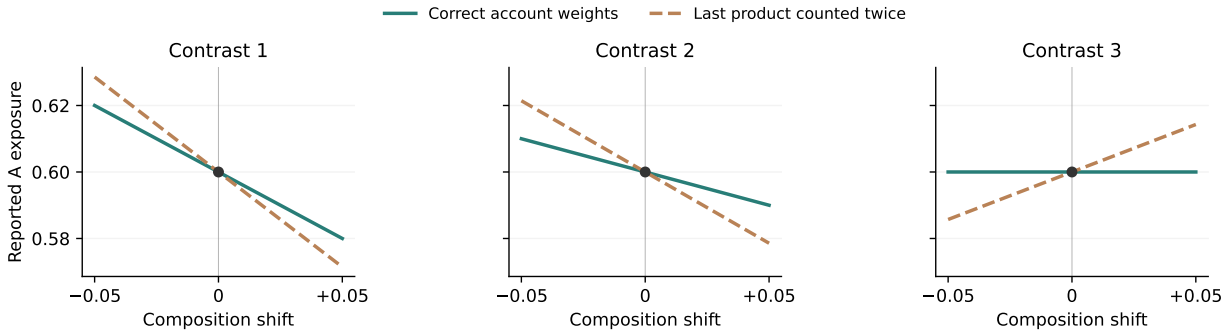


Figure 1: An analytical example with $q = (0.1, 0.2, 0.3, 0.4)$ and homogeneous 60% A holdings. Counting the last product twice gives the same base exposure (black point), but different responses to the three composition contrasts. These lines describe specified linear rules, not observed model outputs.

Corollary 1 (Noise amplification)**.** *If the fixed-linear assumption holds and each queried exposure has additive error of magnitude at most ϵ, then*

$$\|\widehat{p} - p\|_2 \le \frac{\sqrt{3}\epsilon}{2\delta}. \tag{9}$$

Proof. Each centered difference has error at most ϵ/δ. The spectral norm of $C^\top/4$ is $1/2$, yielding the stated bound. □

Small perturbations amplify output noise. The experimental choice $\delta = 0.05$ is fixed before the follow-up evaluation; it is not selected to maximize observed success.

C.2 From weighting distortion to potential risk error

An exposure diagnostic is more informative when its magnitude has an economic interpretation. Let $e = p - q$, and restrict within-product A allocations to $[\ell, u]$. Define the worst-case exposure distortion of the fixed aggregation rule over that composition domain by

$$\mathcal{E}(e; \ell, u) = \sup_{b \in [\ell,u]^J} |e^\top b|. \tag{10}$$

Proposition 3 (Sharp exposure bound and a risk consequence)**.** *For normalized p, q,*

$$\mathcal{E}(e; \ell, u) = \frac{u - \ell}{2} \|e\|_1. \tag{11}$$

For two fixed underlying return series R_A, R_B, the corresponding standard-deviation discrepancy satisfies

$$|\sigma(p^\top b) - \sigma(q^\top b)| \le \sigma(R_A - R_B)\, \mathcal{E}(e; \ell, u). \tag{12}$$

Proof. The positive and negative entries of e have equal total magnitude because $\mathbf{1}^\top e = 0$. Assign $b_j = u$ to positive entries and $b_j = \ell$ to negative entries; this attains $(u-\ell)\|e\|_1/2$, and no admissible assignment can exceed it. For the second statement, write the portfolio return as $R_B + a(R_A - R_B)$. The reverse triangle inequality for the norm of centered random variables bounds the difference of standard deviations by $|a_1 - a_2|\sigma(R_A - R_B)$. □

For the four-product basis, exact contrast measurements therefore give

$$\mathcal{E} = \frac{u-\ell}{8}\|C^\top s\|_1. \tag{13}$$

This score distinguishes a small weighting distortion from one capable of materially changing exposure, even if both give the same answer on the homogeneous base case. If the two assets have identical returns, $\sigma(R_A - R_B) = 0$ and exposure mistakes have no payoff consequence in this restricted model. The risk bound is generally an upper bound, not necessarily attained simultaneously with the sharp exposure bound.

Under Corollary 1, a conservative exposure bound replaces $\|\widehat{e}\|_1$ by $\|\widehat{e}\|_1 + \sqrt{3}\epsilon/\delta$. These are conditional analytical consequences of the fixed-rule model. We do not use them to relabel the empirical audit as a certified market-risk system or retune its frozen acceptance rule.

C.3 Acceptance rule and its limits

The audit uses eight calls: the base description, six centered perturbations, and an anchor with every product entirely allocated to B. For valid fractional outputs, let

$$\kappa = \max_{\ell}\left|\frac{f(b+\delta c_\ell) + f(b-\delta c_\ell)}{2} - f(b)\right|. \tag{14}$$

At the prespecified operating point, the audit accepts the base answer only if

$$\|s\|_\infty \le 0.02, \qquad \kappa \le 0.005, \qquad |f(0)| \le 0.005. \tag{15}$$

Every query must also return valid nonnegative fractions summing to one within 10^{-5}. Missing, malformed, truncated, or invalid outputs cause rejection. The anchor tests an additive offset; curvature tests a local failure of the affine model. The release also reports a tolerance sweep rather than presenting one threshold as universally calibrated.

Even within the fixed-linear class, the empirical slope threshold is not a certificate of the 0.005 accuracy target. For four products, $\|s\|_\infty \le \tau$ implies only $\mathcal{E} \le 3(u-\ell)\tau/4$: the maximum of $\|C^\top s\|_1$ over this cube is 6τ, attained at its vertices. With $[\ell, u] = [0, 1]$ and $\tau = 0.02$, this permits exposure error up to 0.015. The gate is an evaluated operating point, not an analytically calibrated correctness guarantee. Its thresholds remain frozen despite this limitation.

The eight-call implementation is not query-optimal for noiseless linear identification. One-sided contrasts sharing a base query can use fewer calls. We use symmetric pairs to measure curvature and bound difference noise, and a separate anchor to check an additive offset. These additional diagnostics explain the budget; they do not establish universal validity.

Listing 1: Four-product audit. The source structure is supplied by the benchmark.

```
Input: source account weights q, product compositions b
Query: base b; all-B anchor 0;
       b + delta*C[k] and b - delta*C[k], k = 0,1,2
If any answer is invalid or truncated: reject
g[k] = (answer_plus[k] - answer_minus[k]) / (2*delta)
s = g - C*q
curvature = max_k(abs((answer_plus[k] + answer_minus[k])/2
                      - answer_base))
```

```
accept = (max(abs(s)) <= 0.02 and curvature <= 0.005
          and abs(answer_anchor) <= 0.005)
diagnostic_weights = q + transpose(C)*s/4
Return: accept, s, curvature, diagnostic_weights
```

This is not a universal semantic certificate. A language model can change its effective computation between prompts. A nonlinear function can agree with the correct function on any finite set of queried points and disagree elsewhere. Curvature and an anchor reduce particular blind spots but do not remove this impossibility. Reconstructed weights are therefore *conditional diagnostics*, not measurements of internal attention or proof that the model omitted a particular product.

The experiment generates interventions from a known source representation. It does not solve the separate problem of reliably editing arbitrary unstructured financial documents. Moreover, because the experimenter knows q and b, it can calculate the gold answer directly. The audit's role here is to characterize model behavior and benchmark informativeness, not to offer a cheaper replacement for that available calculator.

C.4 An exact-checkable diagnostic of incompatible aggregation

The contrast gate tests whether responses resemble the *correct* weighting rule. Its failure does not establish whether another fixed weighting rule explains them. We therefore add a separately dated, post hoc diagnostic that asks whether *any* nonnegative normalized rule is compatible with the saved answers. This is a bounded-error identification problem in the established set-membership tradition [17]; the optimization uses standard norm approximation and convex duality [4]. The contribution is the portfolio-specific formulation, explicit evidence certificates and empirical diagnosis, not a new linear-programming algorithm.

Let $X \in [0,1]^{T\times J}$ contain the within-product A allocations displayed in the $T = 8$ audit prompts, and let $y \in \mathbb{R}^T$ contain the corresponding reported A exposures. Define

$$d_*(X, y) = \min_{p\geq 0,\, \mathbf{1}^\top p=1} \|y - Xp\|_\infty. \tag{16}$$

This is the smallest uniform answer error needed to reconcile all queries with one admissible aggregation rule. It does not use the prescribed account weights q or the gold aggregate answers. A rule may have $d_* = 0$ and still weight products incorrectly. Conversely, $d_* > \epsilon$ rules out every fixed convex-combination explanation at error budget ϵ; it does not show that every individual answer is wrong or identify a hidden causal mechanism.

Proposition 4 (A verifiable lower bound and an exact dual representation)**.** *For any $\lambda \in \mathbb{R}^T$ with $\|\lambda\|_1 \leq 1$, define*

$$L(\lambda) = \lambda^\top y - \max_j (X^\top \lambda)_j. \tag{17}$$

Then $L(\lambda) \leq d_*(X, y)$, and

$$d_*(X, y) = \max_{\|\lambda\|_1\leq 1} L(\lambda). \tag{18}$$

Any simplex vector $\widetilde{p}$ also supplies the upper bound $d_* \leq U(\widetilde{p}) = \|y - X\widetilde{p}\|_\infty$.

Proof. For any admissible p, the convex-combination property gives $\lambda^\top Xp \leq \max_j (X^\top\lambda)_j$. Therefore $L(\lambda) \leq \lambda^\top(y - Xp) \leq \|y - Xp\|_\infty$ by the dual-norm inequality. Minimizing over p proves the lower bound. For equality, write the infinity norm as a maximum over the unit ℓ_1 ball and interchange the minimum and maximum of the continuous bilinear form on two compact convex sets. Minimizing over the simplex selects the largest coordinate of $X^\top\lambda$. The upper bound follows by evaluating any feasible $\widetilde{p}$. □

The numerical solver proposes primal and dual vectors. We round their coordinates to an integer grid, normalize the primal vector into the simplex, and scale the dual vector to have ℓ_1 norm at most one. A separate checker recomputes $\max(0, L)$ and U using exact rational arithmetic. It requires a bound gap at most 10^{-7}, checks each input against the literal percentages in the archived prompts, and checks each answer against the saved raw response. The checker needs neither an optimizer nor a language model. These are certificates about the printed inputs and saved numerical outputs, not formal verification of the whole software stack.

Simple certificates provide essential comparators. A one-query witness checks whether an answer lies outside the range of its product holdings. A two-query witness checks whether the answer difference lies outside the range of the product-level differences. A three-query midpoint witness checks whether a centered perturbation pair averages to the base response. All are special choices of λ in Proposition 4; the optimizer can combine queries in other ways. The comparison concerns the stated witness families and error budget, not every possible elementary test. In particular, the original slope check already rejects the cases studied below.

We apply the diagnostic to all evaluation cases in both strict and numerically normalized views. A strict case is eligible only when all eight original fractional outputs are valid. The normalized view reuses the previously defined $a/(a+b)$ transformation; it analyzes transformed answers and does not replace primary scoring. The $\epsilon = 0.005$ reference is the existing exposure tolerance, not an estimated noise level. Because this analysis followed inspection of v4 outcomes, it is exploratory, with no additional model calls or independent sample.

C.5 Comparators and a practical computation baseline

The matched eight-call comparator uses the base description, six lot-order permutations, and a version that splits each of four lots into two. It accepts only if all eight outputs are valid and their A exposures span at most 0.005. This measures representation consistency at the same number of calls. Token and latency costs are also reported; matching calls does not guarantee identical computational cost.

We separately evaluate two one-call computation pipelines. *Aggregate extraction* asks the model for account-level A/B weights, then applies a deterministic risk calculator. *Source-factored extraction* asks for one row per funded lot containing its identifier, account weight, and within-product weights. Software validates identifiers, ranges, account-weight conservation, and within-product conservation before calculating $\sum_j \widehat{q}_j \widehat{b}_j$. These checks detect structural failures but cannot guarantee that values were attached to the correct source clauses. The latter pipeline is an established extraction-and-computation baseline, not the novel part of this work.

D Study A: diagnostic design and evaluation

D.1 Recorded pilot and fresh evaluation

The earlier pilot used Qwen2.5-1.5B and Qwen2.5-3B in Q4_K_M form, 16 initial cases, and an eight-case scale replication. Its 472 recorded calls included post hoc follow-ups. The unit illustration copied by the smaller model and the normalization cancellation are reported as motivating failures, with full details in Appendix L. We do not convert post hoc observations into prespecified findings by rewriting the manuscript.

The follow-up contains 18 newly generated families: two for development and 16 for evaluation. Each family supplies one homogeneous and one heterogeneous four-product description with the same aggregate exposure and return distribution. This yields 32 evaluation cases, not 576 independent samples. Account weights are permutations of $(0.10, 0.15, 0.30, 0.45)$ or $(0.05, 0.20, 0.35, 0.40)$. Base A exposure is sampled between 0.28 and 0.72. Heterogeneous within-

product allocations are constructed to preserve that exposure. Three sentence/table grammars vary the order and expression of account versus product weights.

Each case has 18 conditions: base; split lots; six permutations; six perturbations; anchor; source-factored extraction; direct risk generation; and risk generation with verified aggregate weights. Four synthetic joint return states are supplied for risk calculations. They define a descriptive numerical task, not an estimated future market distribution. Risk-changing perturbations are scored on exposure response; risk-generation conditions use the original states and base exposure.

D.2 Models, decoding, and chronology

The fresh evaluation uses Qwen3-8B and SmolLM3-3B, both pinned Q4_K_M artifacts [13, 19]. The release records repository revisions, SHA-256 hashes, runtime command lines, complete API responses, token usage, and request timestamps. Each condition contributes one response to the main evaluation, at temperature zero, with reasoning disabled, JSON-object constrained decoding, and a 512-token output budget. There is no paid API or best-of-n selection. Original observations are never overwritten; the explicitly defined execution recheck below is stored separately. Infrastructure interruptions required resuming requests without a saved response. Reported denominators count saved records rather than every interrupted attempt.

CPU inference uses llama.cpp b12110. Memory pressure required changes to Qwen3 batching, weight repacking and the server's prompt RAM cache. Table 24 reports every actual profile and its saved-response count. These decisions used resource behavior, not answer accuracy; model bytes, prompts, decoding settings and scoring stayed fixed. Recorded segments identify each response's configuration. Batching and kernels can affect numerical reproducibility, motivating a uniform-configuration recheck.

The stimulus and metric protocol was locally hashed before fresh evaluation responses. Analysis implementation was completed while inference ran, following the recorded plan. This is a follow-up designed with knowledge of the earlier pilot, not an externally preregistered study. Four development requests per new model check execution and formatting; they are retained separately and excluded from evaluation accuracy. The larger output budget and removal of illustrative answer values address two specific pilot limitations, while changes in model, prompt, and decoder prevent attributing any cross-version difference to a single factor.

D.3 Uniform-configuration execution amendment

After 360 original Qwen3 core responses had been saved, we froze a separate execution-recheck protocol selecting all 167 observations whose commands differed from the final runtime profile. Selection used request identifiers and commands only, not correctness. Every selected prompt is re-executed once with eight slots, 1,536 context tokens per slot, weight repacking disabled, and the server prompt RAM cache disabled. The full request payload, model bytes and all scoring rules are unchanged.

The main Qwen3 analysis uses these 167 new responses and the 409 original core responses already generated under that profile. Replacement is mandatory for every selected identifier, even if the new response is worse or invalid. The original mixed-profile analysis and paired differences are retained in Appendix O. Source and challenge responses use the final profile and are not repeated. The final ten challenge calls and the recheck resumed after an interruption and workspace reset, with model and engine hashes verified. Uniformity refers to the command profile, not an independently controlled comparison of host hardware or request batches. This is a dated execution amendment, not an independent sample or an externally preregistered replication. It adds 167 records without increasing the 32-case denominator. Across both Study A models, its archived release contains 1,400 original evaluation records, 167 execution-recheck records and

eight separate development records, in addition to the earlier pilot.

D.4 Source-derived holdings tier

Five official Vanguard LifeStrategy factsheets, reporting holdings as of 30 September 2026, provide 48 disclosed fund-level holding rows [23]. Two PDF text extraction engines agree on all 48 name/percentage pairs, and rendered pages are visually checked within the project. No independent human annotation is claimed. Source URLs, page numbers, dates, file hashes, and the transcribed numerical tables are supplied.

For each of the ten fund pairs, constructed account weights of $(0.30, 0.70)$ and $(0.65, 0.35)$ yield 20 scenarios. The task asks for the account fractions invested in three named underlying funds: developed-world ex-UK equity, UK all-share equity, and emerging-market equity. These are a *subset* of account exposures and must not be normalized to sum to one. We compare direct aggregation, split-lot aggregation, and source-factored extraction, giving 60 calls per model.

The published percentages are rounded to one decimal place. We compute gold answers from those displayed values without renormalization. If each published holding is rounded to the nearest 0.1 percentage point, its half-width is 0.0005 in fractional units; a positive weighted average with account weights summing to one has at most the same half-width. The all-target accuracy tolerance is 0.0006. This is a display-precision calculation, not an estimate of market uncertainty. The input is a source-derived holdings table, not an original narrative document or actual client's account. Cases share documents, so we do not treat 20 scenarios as 20 independent financial sources.

D.5 Prospectively frozen challenge extension

During core inference, we added a separate challenge protocol to distinguish rejection of a currently correct answer from prediction of failure on an unqueried composition. An initial version was frozen when 266 of the 576 SmolLM3 core responses had been recorded. Input-only validation found that 26 of its 64 true exposure changes were within the accuracy tolerance. Before any challenge inference, a revised version, frozen after 370 core responses, required a true change of at least 0.01. Both versions are archived. This is a dated exploratory extension, not part of the original core protocol; no challenge model outcomes were used in this revision. Neither gate nor its thresholds changed.

For each of the 32 cases, two new compositions add or subtract $0.04d$, where d is a seeded permutation of $(0.75, -0.35, 0.45, -0.85)$ resampled until $|0.04q^\top d| \geq 0.01$. These directions differ from the three audit contrasts. Challenge queries are unavailable to either eight-call gate. A case passes the challenge only if both outputs are valid and within 0.005 absolute exposure error. We report acceptance coverage, accepted cases that fail this subsequent challenge, and rejected cases that pass it. All cases remain paired by their original family; 64 challenge calls per model are not 64 independent cases. This prospective holdout of compositions tests a narrow form of transfer and does not establish broad model reliability.

D.6 Metrics, uncertainty, and controls

Primary exposure accuracy requires an absolute A-weight error at most 0.005, with structurally valid A/B outputs. Risk accuracy uses a 5% relative-error threshold. Invalid and truncated outputs count as failures. We report audit coverage, correct and incorrect accepted answers, and false rejection of a correct base answer; an audit that rejects everything cannot obtain a useful reliability claim by having zero false accepts.

Paired differences use 10,000 bootstrap resamples of the 16 evaluation families, keeping each

homogeneous/heterogeneous pair together. The intervals are exploratory and quantify variation in this small designed sample. They are not market-risk confidence intervals. Condition-level calls and shared source documents are never counted as independent observations. We report model-specific results rather than pooling checkpoints into a fictional population of models.

Deterministic controls include exact weighted accounting, classical variance computation, and a grammar-specific parser for the controlled templates. Their success does not establish a need for an LLM. Fault injection, described below, is reported separately from language-model outputs. The release preserves all failures rather than replacing them with calculator answers inside a purported model-response file.

E Study A results: identification conditions meet model outputs

E.1 Mechanism validation with injected errors

The executable algebra check enumerates 48 account-weight arrangements and all 80 nonempty vectors in $\{0, 1, 2\}^4$, yielding 3,840 *injected-rule instances*. Of these, 96 have uniform positive multiplicities and preserve effective weights; 3,744 change effective weights. All 3,840 produce exactly the correct homogeneous exposure. For the chosen heterogeneous vector, only the 96 weight-preserving rules produce the exact answer. One contrast detects 3,648 of the 3,744 weighting distortions; the complete three-contrast basis detects all 3,744, with maximum reconstruction error 1.78×10^{-15} in floating-point execution. These are deterministic mechanism checks, not 3,840 model evaluations or estimates of the prevalence of such errors in deployed systems.

E.2 Fresh exposure and risk performance

Main Qwen3 results use the uniform-configuration view described in the execution amendment. All affected responses are replaced according to runtime configuration, never correctness; the original mixed-profile results are reported in Appendix O.

Condition	SmolLM3-3B	Qwen3-8B
Direct aggregate exposure	10/32	13/32
Numerically normalized exposure	13/32	13/32
Split-lot exposure	6/32	6/32
Source-factored exposure	20/32	32/32
Direct risk generation	5/32	2/32
Risk with verified weights	9/32	0/32
Aggregate extraction + calculator	13/32	20/32
Source-factored extraction + calculator	24/32	32/32

Table 2: All-attempt accuracy on 32 cases from 16 paired families per model. Exposure tolerance is 0.005; risk tolerance is 5% relative error. Invalid and truncated outputs count as failures.

For SmolLM3-3B, the paired source-factored minus direct exposure-accuracy difference is 31.2 percentage points (exploratory 95% family-bootstrap interval [9.4, 53.1]). The split-minus-base interval is [-25.0, 0.0] percentage points. The core evaluation has 0 request errors and 0 output-cap terminations in 576 calls.

Among calculated-risk successes for SmolLM3-3B, 3 using aggregate extraction and 4 using source-factored extraction fail the separate exposure-accuracy threshold. Risk and exposure tolerances differ, and risk is not generally an injective function of exposure. These counts demonstrate score disagreement; they do not by themselves identify an exact variance-symmetry mechanism in the observed errors.

For Qwen3-8B, the paired source-factored minus direct exposure-accuracy difference is 59.4 percentage points (exploratory 95% family-bootstrap interval [50.0, 68.8]). The split-minus-base interval is [-37.5, -6.2] percentage points. The core evaluation has 0 request errors and 0 output-cap terminations in 576 calls.

Among calculated-risk successes for Qwen3-8B, 7 using aggregate extraction and 0 using source-factored extraction fail the separate exposure-accuracy threshold. Risk and exposure tolerances differ, and risk is not generally an injective function of exposure. These counts demonstrate score disagreement; they do not by themselves identify an exact variance-symmetry mechanism in the observed errors.

E.3 Audit coverage and incorrect acceptances

Model	Eight-call check	Accepted	Wrong	False reject	Acc. accepted
SmolLM3-3B	Consistency	2/32	0	8	100.0%
SmolLM3-3B	Contrast basis	0/32	0	10	—
Qwen3-8B	Consistency	1/32	0	12	100.0%
Qwen3-8B	Contrast basis	0/32	0	13	—

Table 3: Both checks assess the same base answer. Wrong denotes incorrect accepted base answers; false reject denotes a rejected correct base answer. A dash means no answers were accepted, so conditional accuracy is undefined.

Model	Holdings	Direct	Factored	Audit accepts	Consist. accepts
SmolLM3-3B	identical	9/16	11/16	0	2
SmolLM3-3B	heterogeneous	1/16	9/16	0	0
Qwen3-8B	identical	13/16	16/16	0	1
Qwen3-8B	heterogeneous	0/16	16/16	0	0

Table 4: Stratified descriptive counts. The two holdings types share economic exposure within each family; they are not independent samples.

For SmolLM3-3B, 0/32 cases provide valid outputs on all eight audit queries. Of these, 0 give reconstructed weights outside the admissible simplex tolerance and 0 have base-answer linear-fit error above 0.005. A supplementary source-copy audit finds 18/32 fully faithful ledgers at per-field tolerance 10^{-6}; 2 aggregate-correct ledgers fail that stricter check. This copying check was added during inference and is descriptive, not a frozen primary endpoint.

For Qwen3-8B, 22/32 cases provide valid outputs on all eight audit queries. Of these, 14 give reconstructed weights outside the admissible simplex tolerance and 11 have base-answer linear-fit error above 0.005. A supplementary source-copy audit finds 32/32 fully faithful ledgers at per-field tolerance 10^{-6}; 0 aggregate-correct ledgers fail that stricter check. This copying check was added during inference and is descriptive, not a frozen primary endpoint.

Post hoc all-query normalization. The completed SmolLM3 core run has no case with valid raw fractions on all eight audit queries. After inspecting that result, we added a separately dated diagnostic that renormalizes every query output using the same numerical normalization rule as the base-only baseline. This changes values; it is not format-only cleanup. The thresholds remain unchanged and these results do not replace the primary audit.

After normalization, SmolLM3-3B supplies 32 complete diagnostic vectors: 19 reconstruct inadmissible weights and 20 fail the base-answer linear-fit check. These are compatibility diagnostics

Base tolerance	SmolLM3-3B	Qwen3-8B
0.001	0 / 0	0 / 0
0.005	0 / 0	0 / 0
0.01	0 / 0	0 / 0
0.02	0 / 0	0 / 0

Table 5: Sensitivity analysis: accepted / wrong-accepted counts. The slope threshold is four times the displayed tolerance; curvature and anchor thresholds equal it. Primary correctness labels remain fixed.

Model	Eight valid	Contrast accept/wrong	Consistency accept/wrong
SmolLM3-3B	32	0 / 0	3 / 0
Qwen3-8B	32	0 / 0	1 / 0

Table 6: Explicitly post hoc diagnostic. Correctness is assessed against the normalized base answer. Every existing response is retained, with no threshold tuning or extra model calls.

for the restricted model, not evidence that a particular hidden weighting mechanism has been discovered.

After normalization, Qwen3-8B supplies 32 complete diagnostic vectors: 23 reconstruct inadmissible weights and 19 fail the base-answer linear-fit check. These are compatibility diagnostics for the restricted model, not evidence that a particular hidden weighting mechanism has been discovered.

E.4 Unseen-composition challenge

For SmolLM3-3B, 0/32 cases pass both new composition queries; 10 of the 10 cases with a correct base answer fail at least one challenge query.

For Qwen3-8B, 2/32 cases pass both new composition queries; 11 of the 13 cases with a correct base answer fail at least one challenge query.

Model	Eight-call check	Accept	Fail after	Pass after	Pass but reject
SmolLM3-3B	Consistency	2	2	0	0
SmolLM3-3B	Contrast basis	0	0	0	0
Qwen3-8B	Consistency	1	1	0	2
Qwen3-8B	Contrast basis	0	0	0	2

Table 7: Exploratory transfer test using unchanged core-gate decisions. Fail/pass after means that accepted cases fail/pass the two-query challenge. The queries are withheld from the gate. This is a dated extension designed during core inference, not an independently preregistered benchmark.

E.5 Source-derived holdings results

Source-tier format failures, complete predictions, and per-target errors are retained in the machine-readable analysis. Correctness concerns ownership of named underlying funds, not security-level diversification or future volatility.

Model	Direct	Split lots	Source-factored
SmolLM3-3B	0/20	0/20	7/20
Qwen3-8B	0/20	0/20	19/20

Table 8: All three requested exposures must be within 0.0006 of the displayed-data answer. These 20 constructed accounts reuse five factsheets and ten pairs; no independent-source significance claim is made.

Post hoc unit sensitivity. After both source runs completed, inspection of Qwen3 outputs revealed percentage values in the fractional interface. We therefore separately apply a fixed numerical heuristic: if a target-value block contains a value above one and all entries lie in $[0, 100]$, divide the entire block by 100. For ledgers, apply this per row and separately to the account-weight vector. Conversion decisions do not consult gold answers; other values remain unchanged. This can misinterpret units in other settings and is not format-only cleanup. It leaves the primary results and tolerance unchanged.

For SmolLM3-3B, this changes 0 direct, 0 split and 1 factored responses. The resulting all-target accuracies are 0/20, 0/20 and 8/20, respectively. These exploratory numbers distinguish visible unit failures from the remaining aggregation errors; they do not establish a generally valid repair algorithm.

For Qwen3-8B, this changes 4 direct, 19 split and 1 factored responses. The resulting all-target accuracies are 0/20, 0/20 and 20/20, respectively. These exploratory numbers distinguish visible unit failures from the remaining aggregation errors; they do not establish a generally valid repair algorithm.

E.6 Query and token cost

The two diagnostic gates use eight calls per case, while source-factored extraction uses one. Table 9 reports the observed resource use; Figure 2 summarizes accuracy on the same controlled cases.

Model	Method	Calls/case	Tokens/case	Sum latency (s)
SmolLM3-3B	Contrast basis	8	2617	83.0
SmolLM3-3B	Consistency	8	2773	83.1
SmolLM3-3B	Source-factored	1	522	66.3
Qwen3-8B	Contrast basis	8	2790	315.4
Qwen3-8B	Consistency	8	3000	325.9
Qwen3-8B	Source-factored	1	519	171.6

Table 9: Recorded prompt plus completion tokens and summed latencies per case. This excludes interrupted attempts without a saved response. Calls run concurrently; summed latency is not wall-clock time. Different model profiles, request batches and host load prevent a controlled timing comparison.

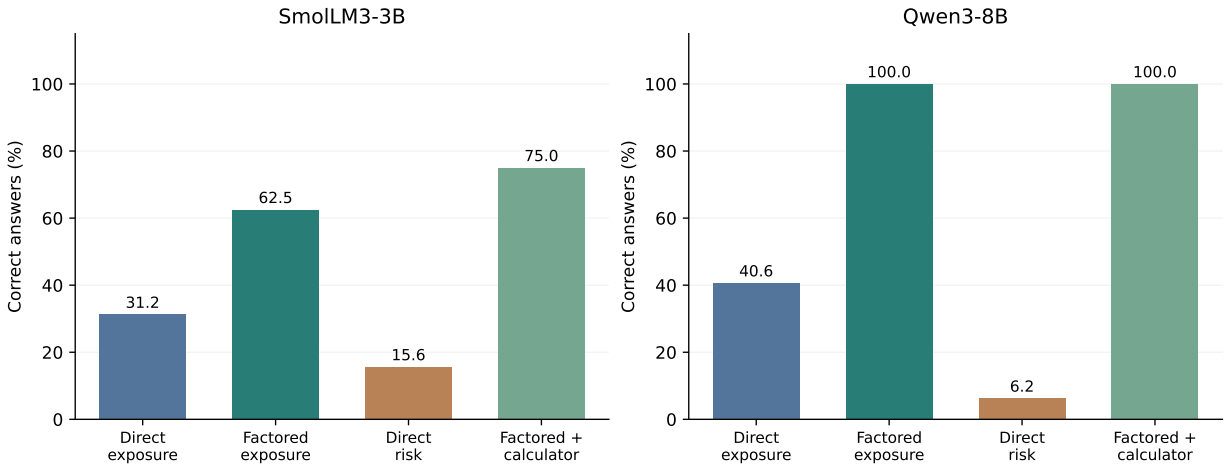


Figure 2: Recorded exposure and risk accuracy. Error bars are omitted because the paired family-bootstrap differences, rather than independent binomial intervals over prompts, are the reported uncertainty analysis.

E.7 Post hoc incompatibility certificates

The full diagnostic and its provenance are specified in Section C.4. No acceptance threshold was retuned. Table 10 distinguishes strict eligibility from numerical normalization; zero eligible cases do not constitute evidence of compatibility.

Model	View	Eligible	Incompatible	Min. d_*	Median d_*
SmolLM3-3B	strict	0/32	0	—	—
SmolLM3-3B	normalized	32/32	32	0.942	7.060
Qwen3-8B	strict	22/32	22	2.066	7.045
Qwen3-8B	normalized	32/32	32	2.066	8.821

Table 10: Incompatible means that the exact lower bound exceeds 0.005. Minimum and median distances are in percentage points of account exposure; displayed values use certified lower bounds, whose upper-bound gaps are below 3.08×10^{-10} in fractional units.

The SmolLM3 strict view supplies no complete eight-query vector. Qwen3 supplies 22, all incompatible at the reference budget. After numerical normalization, all 32 cases from each model remain incompatible: the smallest required uniform error is about 0.94 percentage points for SmolLM3 and 2.07 for Qwen3. Thus the normalized failure cannot be explained by a single fixed normalized multiplicity vector with answer errors bounded by half a percentage point. This rejects a proposed explanation of the recorded responses, not every possible arithmetic or prompt-dependent explanation.

The component analysis also separates formatting from semantic restrictions. After normalization, all 32 cases per model have valid fractions; none passes the original slope criterion. The curvature criterion passes 0/32 SmolLM3 and 1/32 Qwen3 cases, while the anchor criterion passes 28/32 and 32/32. The zero-coverage result is therefore not solely an interface-validity artifact.

Model	One query	Two queries	Midpoint	Union	Full LP
SmolLM3-3B	23	23	31	32	32
Qwen3-8B	21	25	29	31	32

Table 11: Normalized cases rejected by lower-bound witnesses at the unchanged 0.005 budget, out of 32 per model. The union includes the three displayed families. The LP adds no rejected SmolLM3 case and one Qwen3 case; this does not show general superiority over simple tests.

The release contains 86 exact-checked certificates: 22 strict Qwen3 vectors and 64 normalized vectors, with overlapping observations across views. They are not 86 independent cases. Five new software tests include consistent-but-wrong weights, a contradiction with a known exact optimum, and deliberate certificate corruption; the full suite has 18 passing tests. A separate checker also binds each certificate to archived prompt literals and raw outputs. These are internal checks, not independent human validation.

A global contradiction missed by the three simple witness families. The sole additional normalized Qwen3 case is `fresh_13_identical`. Its base and three positive-contrast responses are shown in Table 12. The base answer is correct. Every one-query, two-query and centered-midpoint witness in the specified families has lower bound zero for this case, yet no nonnegative normalized weighting vector fits all responses exactly.

Query	Product A fractions	Reported A
Base	$(0.296, 0.296, 0.296, 0.296)$	0.296
Contrast 1, plus	$(0.346, 0.346, 0.246, 0.246)$	0.246
Contrast 2, plus	$(0.346, 0.246, 0.346, 0.246)$	0.346
Contrast 3, plus	$(0.346, 0.246, 0.246, 0.346)$	0.346

Table 12: Actual saved Qwen3 responses; their original A/B fractions already sum to one, so normalization leaves these values unchanged. This example is the sole case added by the full LP over the stated simple-witness union.

Write the three positive-contrast input rows as x_1, x_2, x_3. Coordinatewise, $-x_1 + x_2 + x_3 = (0.346, 0.146, 0.346, 0.346)$. Any nonnegative normalized weighting rule must therefore satisfy

$$-f(x_1) + f(x_2) + f(x_3) \leq 0.346. \tag{19}$$

The saved answers instead give $-0.246 + 0.346 + 0.346 = 0.446$. If each answer differs from some fixed rule by at most ϵ, the three discrepancies can account for at most 3ϵ of this gap. Consequently $\epsilon \geq 0.100/3$, or $3.\overline{3}$ percentage points. The LP provides a matching upper bound. This is a falsification of an admissible fixed-rule explanation, not proof that the model literally used negative weights. It strengthens the diagnosis of this finite trace; it does not turn the original zero-coverage gate into a useful correctness verifier.

F Study B: matched replication and published holdings

F.1 Controlled design

Study B follows Study A's zero-coverage result. Instead of attempting to certify the model through repeated probes, it tests transfer from structurally uninformative holdings to amount-sensitive holdings and jointly scores each emitted ledger and its total. It is a redesigned follow-up, not a

rerun of Study A: model selection, prompts, output interface, cases and some tolerances change. Only the additional Phi evaluation repeats the frozen Study B controlled prompts. Cross-study accuracy differences are not treatment effects.

The theory and Study A use fractions in $[0, 1]$. Study B displays percentages in $[0, 100]$ and amounts in GBP. For lot i with amount v_i and displayed percentage a_i, its percentage total is $\sum_i v_i a_i / \sum_i v_i$. This is Equation 1 with $q_i = v_i / \sum_k v_k$ and $b_i = a_i/100$, multiplied by 100. Supplied amounts are position values at one snapshot. Ground truth uses exact rational arithmetic, with an additional decimal check for the source extension.

Product	Investment (GBP)	Homogeneous A (%)	Heterogeneous A (%)
1	2,500	55	67
2	1,500	55	27
3	4,000	55	78
4	2,000	55	15
Correct total	10,000	55	55

Table 13: The first frozen controlled family, shown as input rather than a model result. Unweighted product averaging returns 55% in the homogeneous case but 46.75% in the heterogeneous case.

Design. We generated 24 fresh families after earlier exploratory work. Each has unequal product amounts and pairs homogeneous and heterogeneous holdings with the same correct total. The four heterogeneous percentages are distinct, and their unweighted mean differs from the true total by at least five percentage points. This deliberately stresses aggregation and is not representative sampling of customer accounts.

Each account is presented as four funded lots and as six lots after one product is split into 20%, 30% and 50% of its original amount. The split product rotates across families. A genuine-change control moves five percent of the account from the lowest-A product to the highest-A product, changing the correct exposure by at least one percentage point. Four unchanged conditions times two prompt methods, plus one direct change-control request, yield nine requests per family: 216 per model.

Outputs. A direct prompt requests only the total. A separate ledger prompt requests each lot's identifier, amount and within-product target percentage, followed by the model's own total. A deterministic calculator then aggregates that same reported ledger. The arithmetic uses only the reported amounts and percentages; expected lot identifiers from the input are used to check completeness. No values are normalized, rescaled, repaired or imputed. Its result is a third observable output from two model calls, not a third prompting method.

Scoring and uncertainty. A total is correct within 0.5 percentage points. Ledger amounts must match within GBP 0.005 and allocation fields within 0.00005 percentage points. Missing or duplicate lot identifiers and out-of-range values invalidate the ledger. Valid but incorrect ledgers are retained. We separately count correct totals with wrong valid ledgers and exact ledgers with wrong totals. These are output inconsistencies; neither establishes an internal reasoning process.

The primary controlled statistic is homogeneous-minus-heterogeneous direct accuracy, averaging the merged and fragmented contrasts within each family. The original analysis uses 10,000 percentile-bootstrap resamples of the 24 complete families for a 95% interval. The same procedure describes the calculator-versus-direct difference. Related prompts are not treated as independent samples. Intervals are conditional on this generator and fixed configurations, not on the population of financial tasks or models.

Chronology. The controlled prompts, cases, scorer and analysis remain byte-for-byte unchanged between the initial evaluation and replication. We add Phi-4 14B at a pinned third-party Q3_K_S

quantization [1, 16] to the existing Qwen3-8B Q4_K_M results [19]. Replication settings and hashes were locally timestamped before new inference. This is a local prospective record, not external preregistration. All responses remain in the release, including failures.

Run	Quantization	CPU slots	Context per slot
Qwen3-8B, controlled v6	Q4_K_M	8	1,536 tokens
Phi-4 14B, controlled and sources	Q3_K_S	2	2,048 tokens
Qwen3-8B, sources	Q4_K_M	4	2,048 tokens

Table 14: Execution profiles. All use llama.cpp b12110 on CPU, temperature zero, JSON-object output and a 512-token output cap. Qwen thinking is disabled. Different architecture, quantization and execution profiles prevent a causal model-size or quality comparison.

Additional format diagnostic. After the controlled Phi run completed, inspection showed that all 96 ledger responses used the key `lots` instead of the requested `rows`. The frozen primary scorer correctly rejects that interface. To distinguish interface compliance from field fidelity, we added a separately labelled post-hoc sensitivity analysis: when `rows` is absent and `lots` is a list, rename that key alone. No amounts, percentages, lot identifiers or totals change; all other scoring rules remain fixed. The rule is applied to every response in both models and corpora. Five source-Phi responses had already been logged when this rule was defined, so these sensitivity results are not presented as prespecified confirmation. Primary results remain unchanged.

F.2 Extension to published fund holdings

Authentic inputs and constructed accounts. We use the 30 September 2026 Vanguard LifeStrategy Global quarterly report [22] and September 2026 BlackRock MyMap factsheets [3]. Table 15 identifies the eight source tables. There are 88 published holding-name/weight pairs across four PDFs. The target is one exact named underlying fund, not total US equity or geographic exposure.

Parent fund	PDF page	Holding rows	Target weight (%)
VG20: LifeStrategy Global 20% Equity	6	12	0.0*
VG40: LifeStrategy Global 40% Equity	10	16	6.8
VG60: LifeStrategy Global 60% Equity	14	14	14.8
VG80: LifeStrategy Global 80% Equity	18	8	28.3
VG100: LifeStrategy Global 100% Equity	22	8	33.6
MyMap 3, Class S Inc	3	10	13.08
MyMap 4, Class S Inc	3	10	17.89
MyMap 5, Class D Inc	3	10	17.88

Table 15: Vanguard target: Vanguard U.S. Equity Index Fund. BlackRock target: ISHRS US EQ IDX FD (UK) X ACC. *The target is absent from the supplied Vanguard 20% underlying-funds table; zero is an explicit task convention for this named-fund exposure, not a claim of zero US economic exposure.

All ten pairs of Vanguard funds and all three pairs of BlackRock funds form 13 constructed accounts. Each has two unequal investments totaling GBP 10,000. Each account is presented as two lots or as three lots after one investment is split into two parts, with unchanged exposure. Direct and ledger prompts give 52 requests per model. The amounts are generated, not customer records; the fund names and weights are published data. The same printed values determine the oracle, without implying knowledge of unrounded holdings.

Document handling. We extract the selected tables before model inference, standardize them as holding-name/percentage rows, and include both parent funds' tables in each prompt. Thus this tests source-table text, not end-to-end PDF retrieval, OCR, layout understanding or unrestricted narrative interpretation. Every holding-name/weight pair was matched between pypdf and Poppler extraction, and all eight rendered tables were visually compared by the AI assistant. This is a reproducible internal check, not independent human annotation. Source URLs, PDF hashes, page numbers, extracted factual tables and verification records are provided.

Vanguard's underlying-funds tables and BlackRock's Top 10 Holdings tables have different coverage. BlackRock weights are percentages of the whole fund and are not normalized to the top-ten subtotal. The target appears in all three BlackRock tables, so no inference about unlisted holdings is needed. The prompt makes these conventions explicit. It therefore evaluates aggregation under supplied rules rather than discovery of those rules.

Frozen extension and dependencies. Source prompts, accounting answers, scoring thresholds and analysis were saved before any source-model responses. We use the same scoring tolerances as the controlled experiment. All 13 accounts are retained, including the MyMap 4/5 pair whose target percentages differ by only 0.01 points and consequently provide little arithmetic discrimination at the chosen tolerance. Exact absolute errors accompany thresholded accuracy. Accounts reuse eight funds from four documents and two publishers; we report descriptive counts and do not calculate population confidence intervals treating them as independent documents.

A simple exact-match parser is included as a format-specific baseline. It extracts holdings and lots from the standardized prompt text and calculates the total. Its role is to establish how much of this deliberately formatted task ordinary software can solve. Its supported input is the standardized table format, rather than PDFs or unrestricted prose.

G Study B results: transfer, interfaces, and reported fields

G.1 Controlled tasks: 216 requests per configuration

Observable outcome	Qwen3-8B	Phi-4 14B
Homogeneous direct total	48/48 (100.0%)	48/48 (100.0%)
Heterogeneous direct total	5/48 (10.4%)	7/48 (14.6%)
Genuine-change direct total	0/24 (0.0%)	0/24 (0.0%)
Schema-valid exact ledger	96/96 (100.0%)	0/96 (0.0%)
Model total with ledger prompt	55/96 (57.3%)	55/96 (57.3%)
Strict ledger–calculator pipeline	96/96 (100.0%)	0/96 (0.0%)

Table 16: Correct outputs over eligible requests. Totals use the fixed 0.5-point tolerance; exact fields use the stricter ledger tolerances. The 96 ledger requests per model yield both model and calculator totals.

All 96 Phi ledger responses use `lots` instead of `rows`. Its zero strict ledger and pipeline scores therefore reflect interface rejection, not 96 demonstrated numerical-field errors. The format-only sensitivity below evaluates the unchanged numerical content separately.

For Qwen3-8B, the homogeneous-minus-heterogeneous direct accuracy gap is 89.58 percentage points (family-bootstrap 95% interval [81.25, 97.92]). The strict pipeline-versus-direct difference over unchanged cases is 44.79 points [40.62, 48.96].

Heterogeneous direct accuracy is 4/24 (16.7%) for merged lots and 1/24 (4.2%) for fragmented lots. This secondary contrast is descriptive; it does not by itself establish a general fragmentation bias.

For Phi-4 14B, the homogeneous-minus-heterogeneous direct accuracy gap is 85.42 percentage points (family-bootstrap 95% interval [75.00, 95.83]). The strict pipeline-versus-direct difference over unchanged cases is -57.29 points [-62.50, -52.08].

Heterogeneous direct accuracy is 2/24 (8.3%) for merged lots and 5/24 (20.8%) for fragmented lots. This secondary contrast is descriptive; it does not by itself establish a general fragmentation bias.

Under the strict requested schema, among the 96 ledger responses per model, exact fields coexist with an incorrect model total in 41 Qwen and 0 Phi responses. A correct total coexists with a wrong valid ledger in 0 and 0 responses, respectively. These categories should not be conflated.

G.2 Published-table extension: 52 requests per configuration

Observable outcome	Qwen3-8B	Phi-4 14B
Direct total	2/26 (7.7%)	5/26 (19.2%)
Schema-valid exact ledger	23/26 (88.5%)	0/26 (0.0%)
Model total with ledger prompt	5/26 (19.2%)	11/26 (42.3%)
Strict ledger–calculator pipeline	23/26 (88.5%)	0/26 (0.0%)

Table 17: Results on 13 constructed accounts using published holdings. The 26 requests per method reuse eight funds; they are not 26 independent documents.

Phi again uses `lots` in all 26 source ledger responses, so its strict pipeline score reflects interface rejection. The post-hoc format analysis below evaluates the unchanged fields.

Model	Publisher	Direct correct	Direct MAE (pp)	Strict pipeline
Qwen3-8B	Vanguard	1/20	7.264	17/20
Qwen3-8B	BlackRock	1/6	7.120	6/6
Phi-4 14B	Vanguard	3/20	3.496	0/20
Phi-4 14B	BlackRock	2/6	2.019	0/6

Table 18: Descriptive issuer breakdown. Mean absolute error (MAE) uses valid direct totals only; malformed answers remain failures in accuracy. Individual absolute errors and valid-answer counts are retained in the release.

Under the strict requested schema, on source-table ledger responses, exact fields accompany wrong model totals in 18 Qwen and 0 Phi cases. Correct totals accompany wrong valid ledgers in 0 and 0 cases. Both types are reported even when their count is zero.

For Qwen3-8B, source direct totals are correct in 1/13 merged and 1/13 fragmented accounts.

For Phi-4 14B, source direct totals are correct in 2/13 merged and 3/13 fragmented accounts.

As a descriptive check on the test inputs, an unweighted mean of the two parent-fund target percentages passes the fixed tolerance in 1/13 source accounts. This calculation uses no model answers and removes no accounts; it characterizes how many inputs fail to distinguish that particular shortcut.

The format-specific parser obtains 52/52 exact totals and 52/52 exact ledgers from the standardized prompt strings. This is evidence that the supplied table format permits a simple software solution, not evidence of general PDF understanding. It limits any claim that an LLM is necessary for these formatted inputs.

G.3 Post-hoc format-only sensitivity

Corpus	Model	Key aliases	Exact fields	Calculator	Fields right, total wrong
Controlled	Qwen3-8B	0/96	96/96	96/96	41/96
Controlled	Phi-4 14B	96/96	96/96	96/96	41/96
Source	Qwen3-8B	0/26	23/26	23/26	18/26
Source	Phi-4 14B	26/26	26/26	26/26	15/26

Table 19: Secondary, post-hoc analysis after a single key alias. No numeric content is repaired. All primary scores above retain the requested schema.

The alias changes only whether the calculator can access the list. It leaves every model-reported total and every ledger value unchanged. Consequently, an improvement relative to the strict pipeline is an interface effect, not improved model reasoning or a new arithmetic correction. Reported exact fields still do not reveal the internal computation behind either prompt.

A concrete controlled response. For the first frozen family in Table 13, Phi reports all four amounts and product percentages exactly under `lots`, but gives an account total of 47.5%. The same reported numbers yield 55% under the weighted average. Renaming the key changes none of those numbers.

All source-field failures. The three source-ledger field failures after format handling are Qwen responses for VG20_VG40, VG20_VG60, VG20_VG80, all fragmented accounts. Their corresponding merged ledgers are exact. Calculator errors are 2.440, 4.588, 0.589 percentage points, respectively. All three involve the zero-target Vanguard 20% table.

For example, VG20/VG60 should report target percentages [0, 0, 14.8] across its three lots but instead reports [14.8, 14.8, 14.8]. Its model total and ledger-derived total are both 14.8%, whereas the accounting truth is 10.212%. A calculator cannot repair that source assignment. This post-hoc case census includes every source-field failure after format handling.

Execution completeness. Study B contains all 536 planned request records: 432 controlled and 104 source-table requests. The 216 Qwen controlled responses were generated for v6, and 320 further responses were added in v7. The 100 new Study C responses are reported separately in the main paper. There are 0 request errors and 0 length-terminated outputs. Prompt binding, frozen hashes, oracle arithmetic and source-freeze chronology were checked independently of the reporting code.

H Related work and the contribution boundary

Financial text and numerical reasoning. FinQA evaluates numerical reasoning over financial reports using questions and annotated programs [5]. TASER addresses extraction from complex, fragmented holdings tables [6]. Our task begins after document acquisition and table extraction: the models receive controlled narratives or standardized source-table text. It concerns the aggregation and evaluation of supplied holdings, not a solution to unrestricted PDF understanding. PortBench evaluates financial QA and a multi-stage portfolio-management workflow [24]; our accounting task precedes investment choice and does not measure return outperformance.

Behavioral and metamorphic tests. CheckList provides behavioral NLP evaluation, and GSM-Symbolic studies numerical perturbations [18, 20]. Metamorphic relations for NLP are studied systematically by Cho et al. [7]. Investment-menu effects also have a long behavioral-finance history [2, 8]. These precedents mean that neither presentation sensitivity nor perturbation testing is new by itself. Our question is more specific: does a payoff-preserving test actually constrain the product-weighting errors that it is intended to detect? We derive that constraint for a stated error class and then test the interpretation against model outputs.

Calculation and verification. PAL delegates computation to an interpreter [10]. VeriFin grounds numerical claims in filing evidence and checks arithmetic constraints using a solver [11]. Thokal documents the dataset-dependent effects of two fixed scale-correction rules [21]. Our source-ledger calculator follows existing extraction-and-execution practice. We distinguish it from the post-hoc numeric transformations in Study A and the value-preserving key alias in Study B. None is claimed as a new arithmetic algorithm.

Perturbation diagnosis and mathematical tools. ReLoop verifies generated optimization formulations through their responses to parameter perturbations [14]. This is a close precedent for behavioral verification. Algorithm-based fault tolerance, set-membership identification and convex duality also provide established foundations [4, 12, 17]. Our specific construction links an exposure-weighting error class to a complete contrast basis and to checkable incompatible-response certificates. It comes with explicit counterexamples to universal correctness and a failed empirical acceptance policy. The paper makes no first-ever claim about verification, invariance, convex diagnostics or calculator use.

Closest line of work	Established capability	Additional question addressed here
Financial QA / table extraction	Retrieve or report facts and compute answers	Which portfolio test inputs distinguish weighting errors?
Behavioral / metamorphic testing	Check responses to controlled rewrites	Can correct invariance be non-identifying for an entire error class?
Perturbation verification	Detect faults through changed inputs	What rank condition identifies a fixed rule, and do saved LLM traces fit that rule?
Tools and numerical checking	Execute calculations and verify constraints	Is the observed failure in the interface, reported fields, or reported total?

Table 20: Conceptual scope comparison, not a head-to-head performance evaluation or an exhaustive novelty claim.

I Formula provenance and claim ledger

J A worked diagnostic example

Let $q = (0.1, 0.2, 0.3, 0.4)$ and suppose the last product is counted twice while the others are counted once. Then $m = (1, 1, 1, 2)$ and

$$p = (1/14, 2/14, 3/14, 8/14).$$

Component	Provenance and defensible claim
Weighted exposure and variance	Standard accounting and classical portfolio mathematics; not invented here.
Multiplicity model	An analytical error class introduced to study this task; not an observation of hidden model computation.
Homogeneous blind spot	A direct algebraic consequence of normalized aggregation, specialized here to a benchmark-design problem. No priority claim for the algebra.
Rank condition and reconstruction	Standard linear identification and Hadamard-basis identities, applied to the stated exposure audit.
Noise bound	A derived operator-norm bound under bounded additive query errors and fixed weights.
Contrast-basis audit	A task-specific candidate diagnostic with recorded model evaluation. It is not a universal certificate or autonomous document verifier.
Source-factored calculator	Established extraction, validation and arithmetic delegation. It is a practical baseline, not a claimed new reasoning paradigm.
Incompatibility certificates	Standard constrained norm approximation and convex duality, specialized to saved exposure responses; exact arithmetic verifies bounds, not model correctness.
Fault-injection counts	Enumerated artificial error rules, never counted as model responses.
Model results	Derived solely from saved responses, with failures and denominators retained.

Table 21: Separating new study contributions from established mathematics and engineering.

If every product holds 60% A, both $q^\top b$ and $p^\top b$ equal 0.6. The erroneous computation passes the homogeneous exposure test exactly. For the contrast basis in the main text,

$$Cq = (-0.4, -0.2, 0), \qquad Cp = (-4/7, -3/7, 2/7).$$

Their difference is nonzero in every contrast, and Equation 8 recovers $p - q$. This is an analytical example of a specified faulty rule, not an example attributed to an actual model response.

K Why finite probes cannot prove general correctness

Let $\mathcal{Q}$ contain every composition queried by an audit, and let $b^\star \notin \mathcal{Q}$. There exists a function h that is zero at every point of $\mathcal{Q}$ but nonzero at $b^\star$. For example, a sufficiently scaled product of squared distances to the finitely many query points has this property. The function $f(b) = q^\top b + h(b)$ passes all queries in $\mathcal{Q}$ yet disagrees elsewhere. On a compact interior region, scaling can keep the answer within admissible fractional bounds. Therefore the fixed-linear assumption in Proposition 2 is substantive. It cannot be established solely by observing agreement on a finite query set.

Nor do the probes establish the hidden causal mechanism of an LLM. If the model uses different effective weights on different prompts, Equation 8 can return an inadmissible vector or one that does not reproduce the base answer. We retain those compatibility failures rather than describing every reconstructed vector as the model's actual attention or counting pattern.

A further limitation concerns nonzero acceptance tolerances. Set $q = (0.25, 0.25, 0.25, 0.25)$, $p = (0.2635, 0.2455, 0.2455, 0.2455)$ and $b = (0.9, 0.1, 0.1, 0.1)$. Then $s = (0.018, 0.018, 0.018)$, curvature and anchor error are zero, and all ± 0.05 contrasts remain admissible. Thus the specified fixed-linear rule passes the primary gate, while its base exposure error is $0.0108 > 0.005$. This constructed counterexample is separate from recorded model errors. A smaller slope tolerance could imply a stricter bound under the fixed-linear assumption, but changing the threshold after inspecting evaluation results would require a new prospective study.

L Pilot evidence retained for provenance

The pilot remains a distinct, earlier study: 240 core calls and 32 joint probes for Qwen2.5-1.5B, 120 core calls and 16 joint probes for Qwen2.5-3B, and 32 post hoc requests per model, totaling 472. The initial evaluation contains 16 unique economic cases; the second checkpoint uses eight of them. Numerical inputs include both constructed joint states and observed industry-portfolio returns from the Kenneth French Data Library [9]. The holdings narratives and account allocations are constructed in both tiers.

After a format-only cleanup, all 160 recoverable core risk answers from the 1.5B model equal the illustrative value 1.25. Changing that example to 7.91 produces 7.91 in all eight tested risk outputs. Removing the example still does not give an accurate risk answer in that small post hoc set. Among 16 valid joint reports from the 3B model, 15 report the correct aggregate weights, but none of those 15 gives risk within 5% relative error. These observations concern the emitted reports, not hidden reasoning.

Post hoc numerical renormalization increases exact exposure recovery on the 3B homogeneous eight-product subset from 1/8 to 5/8, while the heterogeneous subset remains 0/8. The four additional successes all map $(0.8, 0.8)$ to a balanced $(0.5, 0.5)$ target. That empirical pattern motivates concern about accidental correctness; it does not demonstrate that Equation 4 describes the model's internal mechanism. The new theorem is a separate analytical characterization.

The full earlier manuscript, all 472 raw responses, prompts, answer keys, and analysis scripts are included unchanged in the archived pilot. Its outcomes are not pooled with the fresh evaluation to estimate a single treatment effect. In particular, new models, cue removal, output-budget changes, and constrained decoding are simultaneous design changes, not a clean one-factor replication.

M Source data and rounding

LifeStrategy fund	World ex-UK	UK all-share	Emerging markets
20%	14.4%	4.0%	1.7%
40%	28.8%	8.0%	3.3%
60%	33.2%	12.2%	5.0%
80%	33.1%	16.2%	6.6%
100%	33.0%	20.0%	8.5%

Table 22: Three target underlying-fund weights, as displayed on page 2 of the five issuer factsheets dated 30 September 2026. Complete disclosed holdings tables and source hashes are in the data package.

The displayed fund weights are not silently normalized. The named targets do not exhaust the holdings table. Gold account exposure for target a is $q_1 B_{1a} + q_2 B_{2a}$, using the published percentages divided by 100. For example, allocation to an underlying fund means ownership of that fund through the stated wrappers; it does not imply disjoint stock-level holdings among the targets.

Every source record identifies the official URL, PDF page 2, reporting date, and downloaded-byte hash. Only the numerical transcriptions and retrieval metadata are redistributed in the package; the issuer PDFs remain available at their cited sources. The transcriptions are checked by pypdf and Poppler and by in-project visual inspection. This is not independent human annotation and should not be described as such.

N A recorded transfer failure

The following SmolLM3 example is selected by the first case identifier with a correct base answer and a failed challenge, rather than by maximum error. Case `fresh_03_identical` has fixed account weights $q = (0.2, 0.35, 0.05, 0.4)$. The table displays saved numerical answers to four decimal places; full returned text is retained in the release.

Input	Within-product A weights	True A	Reported A	Reported B
Base	(0.6850, 0.6850, 0.6850, 0.6850)	0.6850	0.6850	0.3150
Plus challenge	(0.6510, 0.7030, 0.6710, 0.7150)	0.6958	0.7515	0.2485
Minus challenge	(0.7190, 0.6670, 0.6990, 0.6550)	0.6742	0.7910	0.2810

Table 23: A recorded correct base answer does not transfer to either new composition. The minus response also violates A/B conservation. These are actual model responses, distinct from the specified faulty rule in Figure 1.

O Execution and analysis contract

Model	Records	Slots	Tokens/slot	Repack	RAM cache (MiB)
Qwen3-8B	61	4	1536	off	8192
Qwen3-8B	64	4	2048	on	8192
Qwen3-8B	700	8	1536	off	0
Qwen3-8B	42	8	1536	off	8192
SmolLM3-3B	700	4	4096	on	8192

Table 24: All raw evaluation and execution-recheck records, including the retained initial mixed-profile observations. The main Qwen3 view uses only the final profile. This excludes eight development responses and interrupted unsaved attempts. HTTP payloads specify cache_prompt=false throughout; zero RAM cache also disables the server-level prompt snapshot cache.

The recheck includes every one of the 167 observations recorded outside the final Qwen3 profile. Returned text differs on 41/167 paired requests; this includes formatting changes and is not an error-rate estimate. The 409 original final-profile core responses are reused. The two views therefore are not independent experiments.

Execution view	Direct	Factored	Direct risk	Audit accepts	Wrong accepts
Initial mixed profile	14/32	32/32	2/32	0	0
Uniform profile	13/32	32/32	2/32	0	0

Table 25: Paired execution sensitivity for the same 32 cases. Both views are retained regardless of which has higher accuracy; the uniform view is used in the main results.

The Study A release separates three evidence types: constructed prompts and gold labels, deterministic fault injections, and raw model observations. A raw observation contains the request identifier, timestamp, payload hash, returned text, full response metadata, token usage, finish reason, and elapsed time. Invalid responses remain in the file. Analysis refuses an incomplete evaluation rather than silently reporting a favorable subset.

The primary fractional-output parser rejects booleans, numeric strings, nonfinite values, negative values, components above one, and nonconservation outside the stated tolerance. The secondary

normalization baseline divides nonnegative finite components by their positive sum. It changes numerical values and is explicitly distinguished from format handling. Risk answers must be finite and nonnegative. Truncation is a failure even if a prefix resembles valid JSON.

Within the archived Study A package, its analysis files are under `experiment_v4/`; the unchanged pilot is under `experiment_v3/`. Reproduce numerical analyses without downloading model weights:

```
python -m unittest discover -s experiment_v4 -p 'test_*.py' -v
python experiment_v4/check_baselines.py
python experiment_v4/audit_release.py
python experiment_v4/verify_certificates.py
python build_results.py
python build_certificates.py
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

The archived README supplies the individual exposure, normalized-output, source, challenge, fault-injection and certificate-analysis commands. The compact sequence above checks saved evidence and rebuilds that historical manuscript; the integrated release has separate commands in Appendix R.

For new inference, use a separate working copy and archive the supplied result directories first. The runner resumes by request identifier and never overwrites an observed response. Refer to the README for pinned downloads, corpus selection, memory requirements, and recorded runtime variations. The model manifests preserve exact revisions and content hashes; a similarly named model at a different revision is not the same experimental artifact.

P Reading the two strict copying failures in Study A

The original supplementary copying check used a per-field tolerance of 10^{-6} in fractional units, whereas aggregate exposure was accepted within 0.005. All cases remain scored under those rules. Two Smol ledger responses pass aggregate exposure but fail field fidelity: `fresh_14_heterogeneous` and `fresh_15_heterogeneous`. These are not two large semantic extraction failures.

In the first, the fourth product's A fraction is reported as 0.5247673, compared with a source value approximately 0.5247763; its B fraction changes correspondingly. In the second, Product 2's B fraction is 0.6578377 instead of 0.6578397, while its A fraction is correctly reported as 0.3421603. The largest discrepancies are about 9×10^{-6} and 2×10^{-6} in fractional units, respectively. These observations illustrate disagreement between differently strict field and aggregate criteria. They do not establish omitted positions, a large weighting distortion, or the homogeneous multiplicity mechanism. This clarification is based on the saved responses, not on new inference; all original counts are retained.

Q Study B source failures and unit conventions

After the value-preserving key alias, every source-field failure is a Qwen response. Table 26 reports all three, without selecting only the largest error. Investment amounts are correctly copied; target percentages are assigned incorrectly.

Split account	True total (%)	Own total (%)	Calculator error (pp)
VG20 / VG40	5.236	10.000	2.440
VG20 / VG60	10.212	14.800	4.588
VG20 / VG80	19.527	22.400	0.589

Table 26: Complete Study B source-field failure census after format handling. The corresponding merged ledgers are exact. Own totals and calculator totals are separate outputs.

The first account should report per-lot target percentages $[0, 0, 6.8]$ but reports $[6.8, 6.8, 1.6]$. The second should report $[0, 0, 14.8]$ but reports $[14.8, 14.8, 14.8]$. The third should report $[0, 0, 28.3]$ but reports $[1.9, 1.9, 28.3]$. The absent-target convention applies only to the exact named underlying fund in the supplied table; it is not a claim of zero geographical US exposure.

Study A source accuracy requires all three target fractions to be within 0.0006. Study B total accuracy uses 0.5 percentage points, equivalent to 0.005 in fractional units. Study B ledger percentages use the much stricter tolerance of 0.00005 percentage points; amount tolerance is GBP 0.005. The studies therefore cannot be ranked by their displayed source success rates. Within each study, all originally frozen tolerances are retained.

R Revision, evidence boundaries and reproduction

The release places the original v8 archive and the completed Study C recheck archive under `evidence/`. The v8 archive in turn contains the immutable v5 and v7 research packages, their historical instructions and test records. Study C records the 100 frozen prompts, both configurations' replies, the scoring program and the two capped responses. The new package verifier checks the source archive digests, raw prompt binding and paired denominators and reruns the frozen scoring in a temporary directory. From the v9 source directory run `python verify.py`. The main paper and this supplement compile separately from `main.tex` and `supplement.tex`.

The recheck resumed after interruptions using saved question IDs. After 69 recorded replies, the experiment moved to a replacement CPU host with matching model-weight and inference-binary hashes. A restored intermediate ZIP contained a stale progress header alongside a newer header. The final archive has one progress header, one raw answer per registered ID, and an audited prefix for the first 69 responses. The final two unsuccessful answers and all analysis choices remain in the archive. Source transcriptions were internally checked with AI assistance. Independent human annotation remains a useful external validation step.

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